

On the ionosphere calibration in GPS radio occultation measurements

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Abstract. Elimination of the ionospheric contribution to atmospheric delays in GPS radio occultation measurements is a key issue in the retrieval of accurate pressure and temperature profiles above the tropopause. The traditional so-called “ionosphere-free” combination of the L1 and L2 phase paths, eliminating the first-order ionospheric effects, is not sufficient because of the nonnegligible bending of the two ray paths. Because of the dispersive nature of the ionosphere the L1 and L2 signals will follow slightly different paths, giving rise to an ionospheric residual, in this paper referred to as the “dispersion” residual. Other higher-order ionospheric effects contribute to the total residual, but the dispersion residual is the most dominant. A linear combination of the L1 and L2 bending angles gives better results in most cases and has in recent years become the method of choice. In this paper a new phase path correction method is presented, dealing with the problem of L1 and L2 ray path separation. Formulas are derived showing how the dispersion residual can be evaluated using measurements of the satellite-to-satellite total electron content. The residuals for various conditions in the ionosphere are estimated numerically and compared to analytic estimates. A mathematical formulation of the difference between this improved phase path correction method and the bending angle correction method is obtained, showing that the two methods are nearly equivalent.

1. Introduction

The Global Positioning System/Meteorology (GPS/MET) experiment, launched in April 1995, has provided accurate temperature and pressure profiles as well as ionospheric electron density profiles using the radio occultation technique [e.g., Ware *et al.*, 1996; Kursinski *et al.*, 1996; Rocken *et al.*, 1997; Hajj and Romans, 1998]. The source of the occultation experiment is the Global Positioning System (GPS) originally designed for precise positioning. In the GPS/MET experiment the GPS signals are received at a low Earth orbit (LEO) satellite, at an altitude of ~750 km. When passing through the Earth's ionosphere and neutral atmosphere, the signals are influenced by the refractive index of the medium. In the geometrical optics approximation the result is that a signal received at the LEO is delayed and has been subject to a small bending (Figure 1). Precise knowledge about the positions and velocities of the satellites (in principle, obtained by simultaneous observa-

tions of different GPS satellites), together with the measured delay, enables the measurement of the bending angle α . Because of the satellite motions the whole atmosphere from top to surface is scanned, obtaining a set of bending angles related to different heights in the atmosphere. Using the assumption of local spherical symmetry, the bending angle profile can be inverted using the Abel transform to yield a refractive index profile [e.g., Fjeldbo *et al.*, 1971].

The refractive index in the atmosphere is related to the ionospheric electron density and to the neutral atmospheric temperature, pressure, and water vapor [e.g., Hartman and Leitingner, 1984]. The electron density in the ionosphere may show large spatial and temporal variations within the region and time frame (~10 min) of an ionospheric occultation, whereas the variations in the neutral atmosphere are more moderate. In the case in which one is only interested in the neutral atmosphere, the scanned region is restricted to the lowest ~100 km of the atmosphere, the occultation lasting ~1 min. During such an occultation the GPS/MET Turbo-Rogue receiver is providing ~3000 measurements at a 50 Hz sampling rate.

Close to the surface the bending angle is ~1–2° depending mostly on the neutral atmospheric contribution to the refraction. As the density in the neutral

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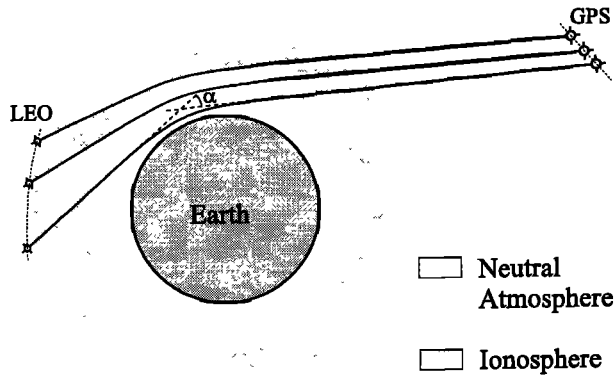


Figure 1. Illustration of the occultation geometry. Signals transmitted by a GPS satellite are refracted by the ionosphere and the neutral atmosphere. When received at a LEO satellite outside the main part of the ionosphere, the signals have been subject to bending.

atmosphere decreases exponentially with height so does the bending angle, but above some 45 km the contribution to the bending from the ionosphere starts overshadowing the contribution from the neutral part of the atmosphere [Hocke, 1997]. At the high-altitude end, the ionospheric contribution dominates, and bending angles may reach 0.015° and 0.025° for the L1 and L2 GPS signals, respectively, corresponding to a ray perigee separation of ~ 500 m [Hardy *et al.*, 1994].

In order to retrieve the neutral atmospheric parameters it is necessary to have an accurate estimate of the bending angle profile due to the neutral atmosphere only. Since the retrieved refractive index profile at some altitude depends on the bending angle profile above that altitude, accurate bending angles to as high altitudes as possible are desirable. As a consequence, occultation measurements of the neutral atmosphere call for efficient methods to eliminate the ionospheric contribution to the signal delay and bending angle.

Transmitting at two frequencies ($f_1 = 1.57542$ GHz and $f_2 = 1.22760$ GHz), the GPS provides a means for correction of ionospheric influences. To first order, the ionospheric refractive index depends on the inverse square of the signal frequency, whereas the refractive index in the neutral atmosphere is independent of the frequency. Thus a simple linear combination of the L1 and L2 signals eliminates most of the ionospheric contribution, resulting in signal parameters which approximately depend on the neu-

tral atmospheric conditions only. Three methods, working on different parameters of the signals, are often seen in the literature:

1. The traditional one, widely used in navigation and geodetic applications of GPS, is the linear combination of the phase paths (or optical path lengths) at the same sample times t [e.g., Spilker, 1980]:

$$L_C(t) = \frac{f_1^2 L_1(t) - f_2^2 L_2(t)}{f_1^2 - f_2^2}. \quad (1)$$

$L_j = \int \mu_j ds$ ($j = 1, 2$, denoting the frequency dependence) is the integral of the refractive index μ along the path from the GPS to the LEO satellite. The dispersive nature of the ionosphere results in the two signals following slightly different paths between the satellites. Besides higher-order ionosphere terms, this gives rise to a residual, in this paper referred to as the “dispersion” residual. Unlike ground-based measurements, the dispersion residual in the occultation geometry has a considerable size, and at daytime conditions during solar maximum the dispersion residual dominates the total ionospheric residual. Since the combination (1) is taken at the same sample times, a similar and equivalent combination can be performed on the excess phases defined as $\Delta L_j = L_j - R_{LG}$, where R_{LG} is the geometrical straight-line range between the satellites at the time in question.

2. Another method, which has been proposed by Vorob'ev and Krasil'nikova [1994] to be used especially for the occultation geometry, is the linear combination of bending angles at a common impact parameter a :

$$\alpha_C(a) = \frac{f_1^2 \alpha_1(a) - f_2^2 \alpha_2(a)}{f_1^2 - f_2^2}. \quad (2)$$

In this correction method, α_j ($j = 1, 2$) is the ray path total bending angle (see Figure 1) at each frequency, as derived either from the Doppler shift [e.g., Fjeldbo *et al.*, 1971] or using the diffraction correction/back propagation method [Karayel and Hinson, 1997; Gorbunov and Gurvich, 1998b]. In a spherically stratified medium the impact parameter is an invariant for a given ray and is defined as the perpendicular distance between either of the ray asymptotes and the center of refraction (see also Figure 2). Vorob'ev and Krasil'nikova originally derived the bending angle correction method using a common ray perigee radial distance as a reference instead of the impact parameter as used in (2).

However, the difference is minor, and the method above has in recent years become the method of choice of ionosphere calibration in GPS occultation measurements [Hocke, 1997; Kursinski *et al.*, 1997; Rocken *et al.*, 1997; Gorbunov and Gurvich, 1998a; Feng and Herman, 1999; Steiner *et al.*, 1999].

3. A third method, applied when occultation data are inverted using Fresnel diffraction theory, is the linear combination of the angular phase modulation at a common ray perigee height h [e.g., Melbourne *et al.*, 1994]:

$$\Psi_C(h) = \frac{f_1 \Psi_1(h) - f_2 \Psi_2(h)}{f_1 - f_2}. \quad (3)$$

Here $\Psi_j = k_j \int (\mu_j - 1) ds$ ($j = 1, 2$) is the atmospheric-induced angular phase modulation at each frequency (k_j being the vacuum wave number) obtained via the Fresnel transform solution. Only the geometrical optics approximation is considered here. Taking into account the Fresnel deconvolution effects related to the thin screen approximation, an additional term should be added to Ψ_j [Melbourne *et al.*, 1994].

The importance of the ray path separation in the GPS occultation geometry and its effects on retrieved atmospheric parameters were first pointed out by Hajj and Kursinski [1991]. They concluded that the traditional dual-frequency combination of phase paths was inadequate at daytime conditions and would be the limiting error source of temperature retrievals in the stratosphere.

The bending angle correction method 2 avoids most of the residual related to the dispersion, because the linear combination is taken at a common impact parameter instead of at the same sample times. The remaining bending angle residual for this method was estimated by Vorob'ev and Krasil'nikova [1994] to be on the order of 10^{-7} rad. Kursinski *et al.* [1997] concluded that at nighttime conditions or at solar minimum this method is sufficient to remove the ionospheric effects to a level below that of other significant errors. However, they also concluded that at daytime conditions, during solar maximum, the ionospheric residual, still remaining after calibration using (2), will be a limiting error source in the reconstruction of atmospheric parameters in the stratosphere and above.

Method 3, like the bending angle correction method, is based on geometrical assumptions, and the

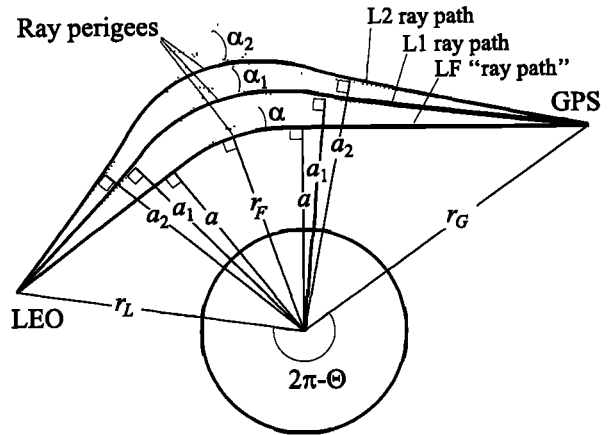


Figure 2. Illustration of the ray path separation and pictorial definition of some parameters used in the text. The L1 and L2 signals follow different paths because of the dispersive ionosphere. LF is the signal path that would have been, had there been no ionosphere.

correction is done at a common altitude in the atmosphere, avoiding most of the dispersion effects.

Several improvements for the ionospheric correction have been proposed over the years. Concentrating on ground-based GPS measurements, Gu and Brunner [1990] and Brunner and Gu [1991] used series expansions to evaluate the dispersion and higher-order effects. However, in order to calculate the residual, knowledge of the gradients of the refractive index along the ray path was needed.

Hardy *et al.* [1994] used the curvature correction [Williams, 1975; Gu and Brunner, 1990] to estimate a "bending" term in the occultation geometry. In a model study, using a representation of a daytime ionosphere in a period of high solar activity, they found that the bending term dominates over other higher-order ionosphere terms and that the error becomes particularly large at an altitude corresponding to the sharp vertical density gradients at the bottom of the *E* layer. They suggested that information on the vertical structure of the ionosphere recovered by inverting the ionospheric occultation data could be used to reduce the errors. However, they also showed that additional information on the horizontal structure of the ionosphere is needed in order to constrain such a solution to obtain an accuracy required for useful correction. Melbourne *et al.* [1994] showed that most of the bending term, which constitutes the first-order dispersion residual at high alti-

tudes, can be evaluated using simplified geometrical considerations.

Gorbunov *et al.* [1996] suggested that the higher-order residuals be corrected for in the bending angle correction method, using a qualified guess of the vertical electron density distribution in the ionosphere. Their simulations showed that the residual in using (2) is most sensitive to the electron density in the lower part of the ionosphere, which is also the region where the ionosphere exhibits the most variations, and therefore is difficult to predict.

Heading toward solar maximum, the need for a better model-independent ionosphere calibration scheme for GPS occultation measurements will become evident. Also, to exploit the potential accuracy of future GPS receivers, it is necessary to find a better correction of the ionosphere than the ones used at present. This paper is the outcome of an attempt to achieve such a goal.

In section 2 a derivation of the residual phase path error resulting when just using method 1 is outlined. The residual is formulated in terms of parameters connected to the satellite-to-satellite total electron content (S-S TEC), which can be obtained from the L1 and L2 phase data themselves during the neutral atmosphere occultation. This results in a new dual-frequency phase path correction method outperforming the traditional method 1. In section 3 the theoretical results from section 2 are validated by numerical simulations, and limitations due to ionospheric asymmetry and lack of a priori knowledge are discussed. In section 4 a few examples of application to real data from the GPS/MET experiment are presented. Based on the results obtained in sections 2 and 3, section 5 outlines the differences and similarities between the improved phase path correction method and the methods 2 and 3 above. Finally, conclusions and remarks are given in section 6.

2. Derivation of the Residual Phase Path Error

This section is devoted to an analytic investigation of the residual phase path error resulting from doing the traditional combination of phase paths (1). All derivations are based on the geometrical optics approximation. Thus it is assumed that the measurements are taken in a region without any sharp vertical gradients or turbulence, so that scintillations, diffraction effects, and multipath propagation can be disregarded. The results, giving a formulation of the

residual phase path error, can then be applied to obtain an improved phase path correction method.

Figure 2 illustrates the L1 and L2 paths together, with the definition of the “ionosphere-free” path, LF, being the path that would have been in the absence of the ionosphere. Ideally, the L1 and L2 phase path observables are

$$L_1 = \int_1 [1 + N_n + N_i(f_1)] ds, \quad (4a)$$

$$L_2 = \int_2 [1 + N_n + N_i(f_2)] ds, \quad (4b)$$

where ds is an element of length along the ray paths. N_n and N_i are the neutral atmosphere and ionosphere contributions to the refractive index, respectively. In this paper these parameters are referred to as refractivities, the total refractivity then being defined as $N = N_n + N_i \equiv \mu - 1$. As mentioned, the ionospheric refractivity N_i is frequency dependent, while for L band frequencies the neutral atmospheric refractivity N_n is not. It is assumed that the problem of phase ambiguities has been resolved and that the observables are free of any noise or residual clock errors. The indices on the integrals indicate that the paths are not identical. The ionosphere-free “observable” is defined as

$$L_F \equiv \int_F (1 + N_n) ds, \quad (5)$$

where the F indicates that the integral is along the ionosphere-free path. L_F is the phase path that we would like to obtain after applying an ionospheric correction based on the L1 and L2 observables. Unfortunately, the combination (1) does not give the required estimate accurately enough in all cases. There will be a difference between L_C and L_F depending on the ionospheric electron density and the Earth’s magnetic field.

Ignoring contributions from positive ions and collisions, the refractive index in the ionosphere can be described by the Appleton-Hartree formula [e.g., Budden, 1985]. Generally, there will be two solutions, splitting the signal into an “ordinary” and an “extraordinary” wave. For the GPS signals, the power of the ordinary wave is only a fraction of the total power of the signal, and the ionospheric refractivity may thus be approximated to second order as [Bassiri and Hajj, 1993]

$$N_i = -C \frac{N_e}{f^2} - K \frac{B_{\parallel} N_e}{f^3}, \quad (6)$$

where f is the carrier frequency, N_e is the electron density (el/m^3) (not to be confused with refractivity), $B_{\parallel}$ is the absolute value of the component of the Earth's magnetic field along the ray path, $C = 40.3 \text{ m}^3/\text{s}^2$, and $K = 1.13 \times 10^{12} \text{ m}^3/\text{Ts}^3$. For GPS occultations, higher-order terms in the expansion of the ionospheric refractivity can be disregarded, also after applying the traditional correction (1) [Hardy *et al.*, 1994; Melbourne *et al.*, 1994]. The second-order term, proportional to the inverse cube of the frequency, results only in a small residual, in practice having very little or no influence even at daytime during high solar activity [Syndergaard, 1999]. For the sake of brevity this term will also be disregarded henceforth.

Now the following nomenclature is introduced:

$$\rho_1 \equiv \int_1 (1 + N_n) ds, \quad \gamma_1 \equiv \int_1 N_e ds,$$

$$\rho_2 \equiv \int_2 (1 + N_n) ds, \quad \gamma_2 \equiv \int_2 N_e ds,$$

$$\rho_F \equiv \int_F (1 + N_n) ds, \quad \gamma_F \equiv \int_F N_e ds.$$

Note that ρ_F and L_F are identical but that it is convenient to operate with separate symbols, since ρ_F is meant as a short notation for an otherwise longer expression, while L_F in (5) is the desired phase path observable. The residual obtained using (1) can now be written as

$$L_C - L_F = \frac{f_1^2(\rho_1 - \rho_F) - f_2^2(\rho_2 - \rho_F)}{f_1^2 - f_2^2} - \frac{C[(\gamma_1 - \gamma_F) - (\gamma_2 - \gamma_F)]}{f_1^2 - f_2^2}. \quad (7)$$

The terms on the right-hand side constitute the dispersion residual ΔL_D . Thus a good approximation for the ionosphere-free phase path may be written

$$L_F = L_C - \Delta L_D. \quad (8)$$

In estimating the two terms on the right-hand side of (7), estimates of the differences between $\rho_j - \rho_F$ and $\gamma_j - \gamma_F$ ($j = 1, 2$) are needed. The idea followed in

this paper is to transform all the involved integrals into the same integration domain by a change of variable. This can be realized using the assumption of local spherical symmetry, so that the neutral atmospheric refractivity, $N_n = N_n(r)$, and the electron density, $N_e = N_e(r)$, are functions only of the radial distance r to the center of refraction.

Only the procedure for calculating $\gamma_1 - \gamma_F$ will be outlined. The calculation of $\rho_1 - \rho_F$ can be carried out using the same principles, but it is far more extensive because many of the calculations require to be carried out to one order higher, and for that reason, only the result will be stated. The results obtained will be directly applicable to $\gamma_2 - \gamma_F$ and $\rho_2 - \rho_F$ just by substituting the subscript 1 for 2.

Using Bouguer's law for a ray path in an isotropic spherically stratified medium [e.g., Budden, 1985] and applying a change of variable into the radial domain, the S-S TEC along the ionosphere-free path may be written as

$$\gamma_F = \int_{r_F}^{r_L} \frac{N_e \mu_n r dr}{\sqrt{\mu_n^2 r^2 - a^2}} + \int_{r_F}^{r_G} \frac{N_e \mu_n r dr}{\sqrt{\mu_n^2 r^2 - a^2}}, \quad (9)$$

where r_F is the perigee radius for the free path (see Figure 2), $\mu_n = 1 + N_n$, and r_L and r_G are LEO and GPS orbit radial distances, respectively. With yet another change of variable, $x = \mu_n r$, (9) can be written

$$\gamma_F = 2 \int_a^{\infty} \frac{x (dr/dx) N_e dx}{\sqrt{x^2 - a^2}}, \quad (10)$$

where it has been assumed that $N_n = 0$ and $N_e = 0$ at the satellite altitudes and above. The first approximation is safe to make since the neutral atmospheric refractivity is effectively zero at heights of the LEO and above. The second approximation, though, may be questioned, especially if the LEO satellite has a relatively low orbit so that it is not effectively free of the ionospheric F layer. The error in doing this approximation is discussed further in section 3. At this point neither of the approximations are strictly necessary, but to simplify calculations later on, similar assumptions will be used, and so it is done here for consistency. Likewise, for the S-S TEC along the L1 path:

$$\gamma_1 = \int_{r_1}^{r_L} \frac{N_e \mu_1 r dr}{\sqrt{\mu_1^2 r^2 - a_1^2}} + \int_{r_1}^{r_G} \frac{N_e \mu_1 r dr}{\sqrt{\mu_1^2 r^2 - a_1^2}}, \quad (11)$$

where r_1 and a_1 are the ray perigee radius and the impact parameter, respectively, for the L1 path, and $\mu_1 = 1 + N_1 = 1 + N_n + N_i(f_1)$ is the complete refractive index for the L1 signal. Using another change of variable, $x = (a/a_1)\mu_1 r$, we get

$$\gamma_1 = 2 \int_a^\infty \frac{x(dr/dx) N_e dx}{\sqrt{x^2 - a^2}}. \quad (12)$$

Now, γ_1 and γ_F in (10) and (12) look alike, but, in fact, they are not. In using different changes of variables to obtain (10) and (12), the functions $N_e[r(x)]$ are different functionals of x , and the same goes for the functions dr/dx . To accurately estimate $\gamma_1 - \gamma_F$, higher-order differences between the numerators of the integrands in (10) and (12) are needed. The goal is to find the difference $\gamma_1 - \gamma_F$ to second order. To achieve this, each of the numerators in (10) and (12) are turned into series expansions (see appendix A), giving the result

$$\begin{aligned} \gamma_1 - \gamma_F = & 2 \frac{C}{f_1^2} \int_a^\infty \frac{x^2 \frac{dN_e^2}{dx} dx}{\sqrt{x^2 - a^2}} + 2 \frac{\Delta a}{a} \int_a^\infty \frac{x \frac{d(xN_e)}{dx} dx}{\sqrt{x^2 - a^2}} \\ & + \left(\frac{\Delta a}{a} \right)^2 \int_a^\infty \frac{x^3 \frac{d^2 N_e}{dx^2} dx}{\sqrt{x^2 - a^2}}, \end{aligned} \quad (13)$$

where $\Delta a = a_1 - a$.

As discussed in appendix A, the recognition of which terms to keep and which to ignore in the expansions was based on estimates using models of the neutral atmosphere and ionosphere. It is important to emphasize that since the models were smooth, the estimates are only valid on the large scale, and consequently, the resulting method cannot be used to calibrate small-scale features. The modeled ionospheric E layer was moderately sharp (see section 3), which should ensure that the evaluations are valid for most large-scale features within the geometrical optics approximation.

Next, each of the integrals in (13) are put into a form more physically comprehensible. In appendix B it is shown how each of the integrals can be simplified using the rule of integration by parts together with Leibnitz's rule for differentiation of integrals. Inserting the results (B1), (B2), and (B3), we can write (13) as

$$\begin{aligned} \gamma_1 - \gamma_F = & \frac{C}{f_1^2} a \frac{d}{da} \int_F N_e^2 ds + \Delta a \frac{d}{da} \int_F N_e ds \\ & + \frac{1}{2} (\Delta a)^2 \frac{d^2}{da^2} \int_F N_e ds. \end{aligned} \quad (14)$$

An approximate expression of Δa in terms of $\int_F N_e ds$ and its derivatives is derived in appendix C. By substituting (C9) into (14), defining

$$\beta_F \equiv \int_F N_e^2 ds, \quad (15)$$

and revoking the definition of γ_F , we obtain the expression for $\gamma_1 - \gamma_F$ to second order:

$$\begin{aligned} \gamma_1 - \gamma_F = & D \zeta \frac{C}{f_1^2} \left(\frac{d\gamma_F}{da} \right)^2 + a \frac{C}{f_1^2} \frac{d\beta_F}{da} \\ & + \frac{3}{2} (D \zeta)^2 \frac{C^2}{f_1^4} \left(\frac{d\gamma_F}{da} \right)^2 \frac{d^2 \gamma_F}{da^2}. \end{aligned} \quad (16)$$

In (16), D is the reduced distance, and ζ is the defocusing factor as defined in appendix C.

The last term in (16) is obviously the higher-order term, being inversely proportional to the fourth power of the frequency. However, at impact altitudes of interest here, the second term on the right-hand side turns out to have a relatively small magnitude also, which will be elaborated in section 3.

As mentioned, the derivations leading to an expression of $\rho_1 - \rho_F$ to second order will not be given here. The calculations are somewhat similar, but manyfold more extensive because ρ_1 and ρ_F themselves have lower order than γ_1 and γ_F , and higher-order series expansions are therefore needed. With a lot patience it is possible to obtain

$$\begin{aligned} \rho_1 - \rho_F = & \frac{1}{2} D \zeta \frac{C^2}{f_1^4} \left(\frac{d\gamma_F}{da} \right)^2 + \frac{1}{2} a \frac{C^2}{f_1^4} \frac{d\beta_F}{da} \\ & + (D \zeta)^2 \frac{C^3}{f_1^6} \left(\frac{d\gamma_F}{da} \right)^2 \frac{d^2 \gamma_F}{da^2}. \end{aligned} \quad (17)$$

Note that $\rho_1 - \rho_F$ depends on basically the same terms as $\gamma_1 - \gamma_F$; only the coefficients are different. Similar expressions to (16) and (17) hold for $\gamma_2 - \gamma_F$ and $\rho_2 - \rho_F$ just by substituting the frequency. Combining it all into the terms on the right-hand side of (7), we finally obtain the formula for the dispersion residual:

$$\Delta L_D = \frac{1}{2} \frac{C^2}{f_1^2 f_2^2} \left[\Gamma + C \left(\frac{1}{f_1^2} + \frac{1}{f_2^2} \right) \Lambda \right], \quad (18)$$

Γ and Λ being defined as

$$\Gamma \equiv D\zeta \left(\frac{d\gamma_F}{da} \right)^2 + a \frac{d\beta_F}{da}, \quad (19)$$

$$\Lambda \equiv (D\zeta)^2 \left(\frac{d\gamma_F}{da} \right)^2 \frac{d^2\gamma_F}{da^2}. \quad (20)$$

The part of the residual proportional to Γ contains two separate terms. The term depending on the square of the S-S TEC (γ_F) gradients will be referred to as the “major” dispersion term, while the term depending on the gradients of the integrated squared electron density (β_F) will be referred to as the “minor” dispersion term. The reasoning for this terminology will become evident in section 3. The sum of the major and the minor dispersion terms is the first-order dispersion term. The term proportional to Λ will be referred to as the second-order dispersion term.

The major dispersion term constitutes the major part of the dispersion residual, and hence the formulas (18)–(20) show that the dispersion residual is mainly proportional to the square of the S-S TEC gradients.

In principle, the formulas (18)–(20) together with (1) and (8) constitute an improved dual-frequency phase path correction method for the ionosphere calibration of GPS occultation measurements. However, some modification and discussion, which follow in section 3, are needed before it can be applied in practice.

3. Simulations

3.1. Numerical Validation

The theoretical results obtained in section 2 were validated by numerical estimation of the residuals using ray tracing. In the estimations the ionosphere E and F layers were represented by the superposition of two Chapman profiles,

$$N_e = N_e(h_0) \exp \left\{ \frac{1}{2} \left[1 - \frac{h - h_0}{H} - \exp \left(- \frac{h - h_0}{H} \right) \right] \right\}, \quad (21)$$

where $N_e(h_0)$ is the maximum electron density, h is the height above the Earth's surface, h_0 is the height

of maximum electron density, and H is a scale height. To simulate the conditions of a typical daytime ionosphere at solar maximum, the following parameters for the E and F layers were chosen. For the E layer, $N_e(h_0) = 2 \times 10^{11} \text{ m}^{-3}$, $h_0 = 105 \text{ km}$, and $H = 5 \text{ km}$. For the F layer, $N_e(h_0) = 3 \times 10^{12} \text{ m}^{-3}$, $h_0 = 300 \text{ km}$, and $H = 60 \text{ km}$. The existence of the $F1$ layer and the D layer was not considered, because the main effect on neutral atmosphere occultations comes from the bulk ionosphere, represented by the F layer, and the sharp gradients at the bottom of the ionosphere, represented by the E layer. The E layer scale height was chosen to be moderately small (5 km), more or less simulating the sharpness of a sporadic E layer, though still staying within the limits of the geometrical optics approximation, avoiding multipath propagation.

The simulation of the neutral atmospheric refractivity was based on the Mass Spectrometer Incoherent Scatter Model–Extended (MSISE-90) [Hedin, 1991]. A simplified version of that model was parameterized for fast ray-tracing purposes to give the refractivity as a function of height, latitude, longitude, and month. In the present study only the height dependence in the parameterized model was used.

The calculation of residuals was based on three-dimensional (3-D) ray tracing using the canonical ray path equations in an isotropic medium [Budden, 1985]. The ray-tracing algorithm was developed at the Danish Meteorological Institute in connection with a study for the European Space Agency [Høeg *et al.*, 1996] and has shown to have an accuracy better than 0.05 mm. The algorithm simulates radio occultations, utilizing real GPS and LEO orbit ephemerides from the GPS/MET experiment.

The dispersion residual was calculated in two different ways. First, an occultation was simulated using only the model of the neutral atmosphere, i.e., without the ionosphere. The synthetic data obtained in this way are the “true” ionosphere-free phase path L_F as a function of time during the occultation. In the same process the trajectories for all the ray paths were calculated at sufficiently many points. Then, using the ionosphere model, the electron density and the square of the electron density were integrated along the LF paths, obtaining γ_F and β_F . The impact parameters a and the bending angles α were calculated from the L_F data using standard techniques [e.g., Melbourne *et al.*, 1994]. The necessary differentiations with respect to the impact parameter were

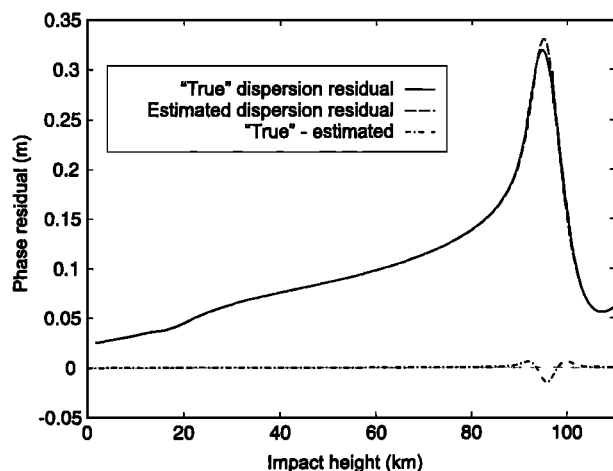


Figure 3. Comparison of “true” (solid curve) and estimated (dashed curve) dispersion residuals as a function of the impact height (the impact parameter minus Earth’s radius) and the difference between them (dash-dotted curve).

performed numerically to finally obtain the dispersion terms in (19) and (20).

A second simulation was done, now including both the ionosphere and the neutral atmosphere models in the ray tracing. This simulation gave the time-tagged phase path observables, L_1 and L_2 , which, combined in the traditional phase path correction using (1), gave the ionosphere-corrected phase path L_C . The difference between L_C and L_F is the true dispersion residual. The residual obtained using (18)–(20) is the estimated dispersion residual.

Figure 3 shows the true and estimated dispersion residuals and the difference between them. The dispersion residual is shown to reach its maximum of several tens of centimeters at an altitude of 95 km, ~10 km below the E layer peak in this example. This corresponds to the region of maximum S-S TEC gradients, being a little lower than the height of maximum vertical gradients of the electron density profile. The S-S TEC gradients are mostly controlled by the local vertical electron density gradients at the perigee height (for the range of concern here, this means the bottom E layer gradients) but also to a lesser extent by the bulk ionosphere (the F layer gradients) which the rays have passed through farther away from the ray perigee. The difference between the true and the estimated dispersion residual is very small, and it only becomes appreciable close to the height where the dispersion residual is a maximum.

This remaining part is primarily due to a third-order dispersion term, not taken into account in the derivations. Such a term would depend on the third derivative of the S-S TEC with respect to the impact parameter a .

To investigate further the contributions from each of the dispersion terms to the total estimated dispersion residual, the major, the minor, and the second-order dispersion terms were estimated and are shown in Figure 4. The major dispersion term has clearly the most influence, especially because it is the derivatives of the residuals, and not the residuals themselves, that determine the bending angle residual. In that respect, above ~80 km the second-order term becomes more dominant than the minor term. However, the minor term has an almost linear trend at all altitudes, which will map into a bending angle bias. Since the bending angle profile, because of the neutral atmosphere, will decrease nearly exponentially with height, such a bias will normally be the limiting error at high altitudes before the second-order dispersion term takes over. These considerations should be seen in the light of that the major and the second-order dispersion residual terms, as shown in section 3.2, can be estimated from the occultation data themselves and thereby eliminated, while the minor term needs auxiliary information to be estimated accurately.

A Chapman profile representation of the ionosphere as used in the simulations in this study is known to fall off more rapidly with height above the

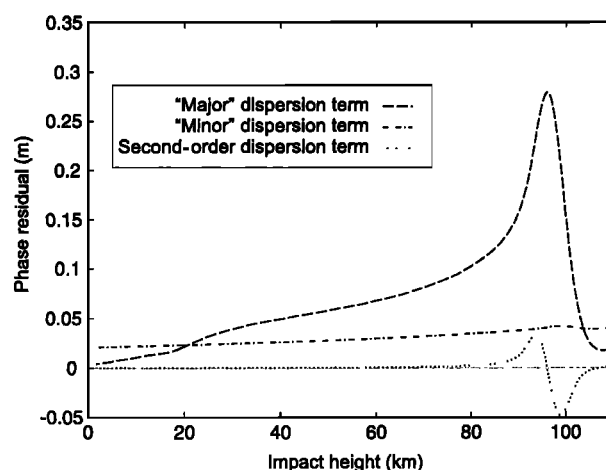


Figure 4. The “major” dispersion (dashed curve), the “minor” dispersion (dash-dotted curve), and the second-order dispersion (dotted curve) terms as a function of the impact height.

F layer than the real ionosphere. In the derivation of the dispersion residual in section 2 the electron density at the LEO satellite (~ 750 km) and above was ignored, which is a good approximation using the Chapman profile but may not be good enough when it comes to the real ionosphere. For a LEO satellite in a much lower orbit than the one modeled here, the approximation is not valid. The error of neglecting the electron density at and above the LEO is difficult to assess, since this approximation is done at several stages of the derivations in section 2. Using an exponentially decreasing model for the electron density, the error in the bending angle (essentially the error we make in (C4)), ignoring the electron density at the LEO, was estimated by *Hajj and Romans* [1998] to be $<0.5\%$. However, the relative error at other stages of the derivations may be even larger. The error made in (10) could be several percent depending on how the electron density falls off with height above the LEO. On the other hand, when expressing the final result in terms of the integrated electron content between the satellites (e.g., when going from (13) to (14)), most of the error probably cancels out, but it would require a closer inspection to find out how much.

3.2. Calculating the Major and the Second-Order Dispersion Terms

In section 2, γ_F and β_F were defined as $\int_F N_e ds$ and $\int_F N_e^2 ds$, respectively. In practice, γ_F can be estimated using the traditional measure of the S-S TEC [e.g., *Sardón et al.*, 1994]:

$$I = \frac{f_1^2 f_2^2}{C(f_1^2 - f_2^2)} (L_1 - L_2). \quad (22)$$

However, since this measure, like the traditional ionosphere correction, (1), is also affected by ray path separation, it is not sufficient to just replace γ_F with I in (19) and (20). An additional second-order effect will appear in (19). Noting that to first order (ignoring second- and higher-order dispersion and ionosphere terms),

$$I = \gamma_F + \frac{1}{2} C \left(\frac{1}{f_1^2} + \frac{1}{f_2^2} \right) \Gamma, \quad (23)$$

it can be shown that to concordant order (ignoring derivatives of D and ζ),

$$\Gamma = D\zeta \left(\frac{dI}{da} \right)^2 + a \frac{d\beta_F}{da} - 2C \left(\frac{1}{f_1^2} + \frac{1}{f_2^2} \right) \Lambda, \quad (24)$$

$$\Lambda = (D\zeta)^2 \left(\frac{dI}{da} \right)^2 \frac{d^2 I}{da^2}. \quad (25)$$

The parameters D , ζ , and a are less sensitive to ionospheric effects and may be calculated with sufficient accuracy using either the traditional phase path correction method or the bending angle correction method.

It may be convenient to keep the first- and the second-order effects as separate terms, and so we may define new symbols, $\tilde{\Gamma}$ and $\tilde{\Lambda}$, indicating calculated measures using the available occultation data:

$$\tilde{\Gamma} = D\zeta \left(\frac{dI}{da} \right)^2 + a \frac{dJ}{da}, \quad (26)$$

$$\tilde{\Lambda} = -(D\zeta)^2 \left(\frac{dI}{da} \right)^2 \frac{d^2 I}{da^2}. \quad (27)$$

Note the change of sign in (27) compared to (25). In (26), J has been introduced symbolizing a calculated (or modeled) counterpart to β_F in the minor dispersion term. $\tilde{\Gamma}$ and $\tilde{\Lambda}$ will then take the place of Γ and Λ in (18). A suggestion for modeling the minor dispersion term is discussed in section 3.3.

3.3. Modeling the Minor Dispersion Term

For ground-based measurements it has been shown [*Hartman and Leitingner*, 1984; *Brunner and Gu*, 1991] that there is a near proportionality between $\int N_e^2 ds$ and $\int N_e ds$. This proportionality can be expressed through a shape parameter,

$$\eta = \frac{\int N_e^2 ds}{N_m \int N_e ds}, \quad (28)$$

N_m being the maximum electron density of the bulk ionosphere (the F layer peak). For vertical incidence in ground-based applications the shape parameter for a Chapman layer model is 0.680 [*Hartman and Leitingner*, 1984]. *Brunner and Gu* [1991] used a value of 0.66 which was estimated for a modified Chapman layer model, and they found only little variation with the elevation angle of the satellite-ground link.

For occultations, referring the integrals to the ionosphere-free path, the shape parameter was found in this study to vary only slightly with the impact height in the range 0–110 km. For a variety of the E and F layer parameters, $N_e(h_0)$, h_0 , and H , using the double-Chapman layer model, the shape parameter

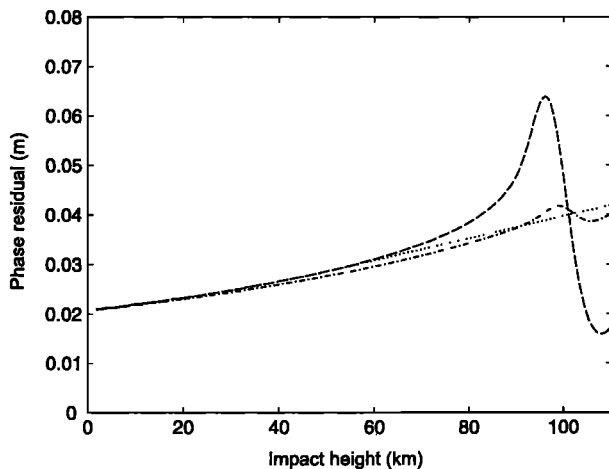


Figure 5. The estimated minor dispersion term as a function of the impact height (dash-dotted curve) and the approximation (dashed curve) using (29), which is valid only at low impact heights. A linear extrapolation above 50 km is shown as the dotted curve.

was between 0.62 and 0.69 in that height range. However, what we need here, in order to model the minor dispersion term, is the derivative of $\beta_F = \int_F N_e^2 ds$. A similar relation between $d\beta_F/da$ and $d\gamma_F/da$ was found to be less constant at high altitudes, primarily due to the *E* layer bulge. A near proportionality between the two was found to be valid only for low impact heights. Fortunately, as seen from Figure 4, the minor dispersion term shows no large variations, and a linear extrapolation from a model being valid at low altitudes may be used as an approximation at higher altitudes.

The maximum electron density N_m may be observed by digisondes from ground-based stations, or, in general, it can be extracted from a global ionosphere model. An alternative estimate is $I_\perp = \tau N_m$, $I_\perp$ being the total electron content in a vertical column from the Earth's surface and up. The relation between N_m and $I_\perp$ involves an "equivalent slab thickness" τ , which does not depend drastically on different ionosphere conditions [Hartman and Lettinger, 1984]. Global maps of the vertically integrated total electron content can be produced on the basis of the Global Positioning System with an accuracy of ~ 2 – 3 total electron content units (TECU) ($1 \text{ TECU} = 10^{16} \text{ el/m}^2$) [Wilson *et al.*, 1995; Jakowski, 1996].

To model the minor dispersion term, one may use the following approximate relation below 50 km:

$$\frac{d\beta_F}{da} \approx \frac{\eta}{\tau} I_\perp \frac{d\gamma_F}{da}. \quad (29)$$

As an example, values of η and τ were set to 0.7 and 250 km, respectively, and $I_\perp$ was calculated numerically from the double-Chapman layer model to be ~ 75 TECU. Above 50 km a linear extrapolation was used based on the data points of the approximation (29) between 45 and 55 km. Figure 5 shows how the minor dispersion term and the approximation are quite close at low impact heights, while higher in the atmosphere, the *E* layer affects $d\beta_F/da$ and $d\gamma_F/da$ differently. However, using the linear extrapolation above 50 km, the phase trend of the minor dispersion term is modeled more accurately all the way up to ~ 80 km. Uncertainties in the values of η , τ , and $I_\perp$ will map almost linearly into the model of the minor dispersion term and its slope.

Of course, if one has a more precise knowledge about the electron density distribution in the ionosphere, i.e., a reliable model, ray tracing may be performed to estimate the minor dispersion term.

3.4. Nonspherical Symmetry

So far we have only discussed the spherical symmetry case. The derivation and the estimation of the dispersion residual were based on the spherical symmetry assumption, and so were the estimation of the shape parameter and the modeling of the minor term in section 3.3. The question one may ask is, how well do these estimates hold when the ionosphere exhibits large horizontal variations over the range in which the occultation rays penetrate the ionosphere? To test the validity of (18)–(20) in such a case, numerical simulations with an asymmetrical ionosphere, representative of a day-night or night-day transition in the region of the occultation, were performed. The asymmetry was modeled using the double-Chapman layer model, gradually decreasing the *E* and *F* layer peak densities toward the "night" region. The situations for each of the two occultation simulations, "day-night" and "night-day" asymmetry, are shown in Figure 6. The rays in the simulation were approximately going west-east (with a small south-north component), and the longitude of the perigees was at $\sim 10^\circ\text{E}$. Estimated residuals, obtained using (18), (19), and (20), are compared with the true residuals, calculated as $L_C - L_F$ (Figure 7). The scale is the same as in Figure 3, giving a clear picture of the much smaller size of the residuals compared to the daytime symmetry case.

The shape parameter was found to be significantly larger in both asymmetrical cases (close to 1.0) and generally decreasing with increasing impact height (a few percent in the range 0–110 km). Anyway, the slope of the difference (dash-dotted curves in Figure 7) becomes comparable with the slope of the minor dispersion term, making it insignificant whether to try to model the minor dispersion term or just leave it. Both strategies will result in a small (but different) bending angle bias. In fact, as will be shown in section 5, the bending angle obtained by setting $dJ/da = 0$ in (26) is almost identical to the bending angle obtained using the ionosphere bending angle correction method (2).

4. Application to Real Data

In summary, the dual-frequency ionosphere correction of the phase path may be improved beyond the traditional phase path correction, taking into account the dispersive influences. In practical applications the dispersion effect may be estimated using measurements of the S-S TEC, I , and information on the vertical TEC, $I_{\perp}$, at the occultation site. The impact

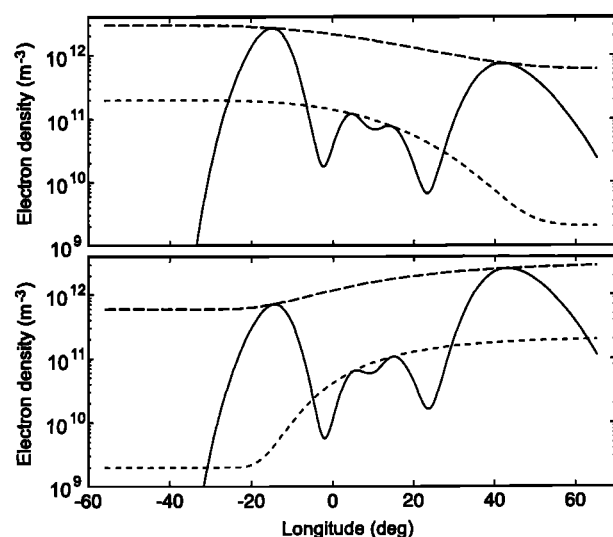


Figure 6. Electron densities as a function of longitude for the modeled occultations. (top) “Day-night” asymmetry. (bottom) “Night-day” asymmetry. The F layer peak density (long-dashed curves) decreases by a factor of 5 over the longitude span of $\sim 60^\circ$ (corresponding to ~ 3000 km at the latitude where the simulated occultation occurs), while the E layer peak density (short-dashed curves) decreases by a factor of 100. The solid curves are the electron densities experienced by a ray having a perigee altitude of ~ 100 km.

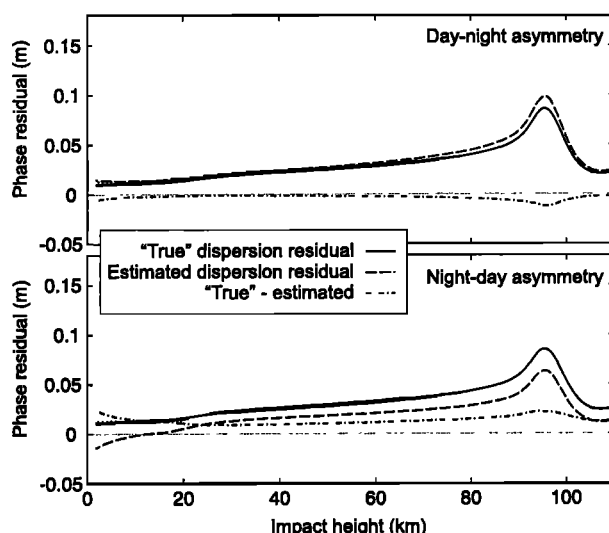


Figure 7. True (solid curves) and estimated (dashed curves) dispersion residuals as a function of the impact height in the two ionospheric asymmetry cases. The difference is shown as the dash-dotted curve.

parameter a , the reduced distance D , and the defocusing factor ζ may all be estimated with sufficient accuracy using either the traditional phase path correction method or the bending angle correction method.

As an example, the phase data from a real GPS/MET occultation were calibrated using both the traditional phase path correction method and the improved method. As with the traditional method, the improved correction may be applied to the excess phase path as well as to the total phase path, since the only difference is a constant term common for both the L_1 and L_2 phase paths at each sample time, which will not affect $L_1 - L_2$ differences used in the determination of the S-S TEC. Figure 8 shows the excess phases as a function of time during the occultation. Only the uppermost part of the occultation, corresponding to heights above 50 km, is shown. At lower altitudes the excess phase becomes very large and may reach 1000 m near the surface. The absolute level on the ordinate is off because of the arbitrary setting of the first point in the data set (equals 1 m for University Corporation for Atmospheric Research (UCAR) data). Ideally, the ionosphere-corrected excess phase should approach 0 m asymptotically at the high-altitude end. However, because of the phase ambiguity in GPS phase measurements the absolute level of the phase is not known a priori. Fortunately,

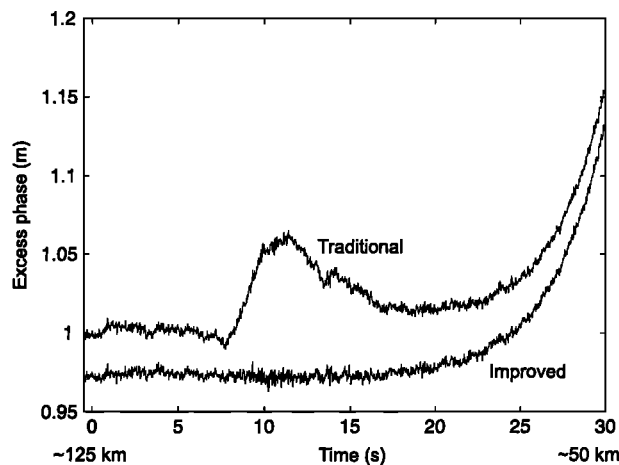


Figure 8. Ionosphere-corrected excess phases as a function of time, using either the traditional phase path correction method or the improved method. The data are taken from the GPS/MET experiment (University Corporation for Atmospheric Research (UCAR) data, occultation 259, day 35, 1997). A value of 40 TECU was used for the vertical TEC in the estimate of the minor dispersion term.

ambiguity determination is not needed in order to find the bending angles from the phases, because all the information needed lies in the phase change. For the same reason, measurements of the S-S TEC do not have to be ambiguity-resolved, since everywhere they are used in the improved method, derivatives are taken.

In Figure 8 a clear bulge due to the *E* layer gradients is seen in the result using the traditional correction method. This bulge is completely removed using the improved method. Because of the high-frequency noise in the real data, all derivatives, calculated numerically, were taken with an interval of 0.6 s between differencing points. The GPS/MET data are from a period of low solar activity (the $F_{10.7}$ solar flux index was 81 for this particular day), and therefore the residual bulge in this example, resulting from the traditional phase path correction, is much less than the residual modeled above (Figure 3). In a period of high solar activity, much larger dispersion residuals are expected.

It should be emphasized that this example was particularly chosen because there are no scintillations which would violate the geometrical optics approximation on which the derivations in section 2 are based. Another example of a GPS/MET data calibration is shown in Figure 9 where large scintillations are still not calibrated very well with the improved

method. However, most of these phase oscillations are at such high altitudes (the peak in Figure 9 is at ~ 100 km) that they appear on the “flat” part of the curve, beyond the altitude where the noise overshadows any signal from the neutral atmosphere manifested as the large-scale slope of the calibrated excess phase. In the lower part of the atmosphere, diffraction and multipath effects, primarily due to water vapor concentrations, may give such large fluctuations that the improved correction, involving numerical differentiations, cannot be applied without considerable smoothing of the L1 and L2 excess phases.

5. Comparison of Methods

The analytical approach, leading to estimates for the dispersion residual for the phase path correction method, can also be applied to give the residual error for the bending angle correction method. For a common impact parameter a we may write

$$\alpha(a) = -2a \int_{r_F}^{\infty} \frac{(d \ln \mu_n / dr) dr}{\sqrt{\mu_n^2 r^2 - a^2}}, \quad (30)$$

$$\alpha_1(a) = -2a \int_{r_1}^{\infty} \frac{(d \ln \mu_1 / dr) dr}{\sqrt{\mu_1^2 r^2 - a^2}}, \quad (31)$$

where $\alpha(a)$ is the bending angle of the ionosphere-free path for a given impact parameter and $\alpha_1(a)$ is

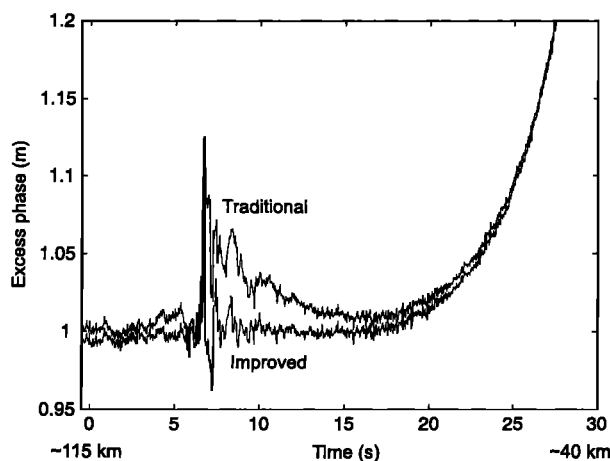


Figure 9. Ionosphere-corrected excess phases as a function of time, using either the traditional phase path correction method or the improved method. The data are taken from the GPS/MET experiment (UCAR data, occultation 67, day 35, 1997). A value of 30 TECU was used for the vertical TEC in the estimate of the minor dispersion term.

the bending angle of the L1 path having the same impact parameter but a slightly different perigee radius, r_1 . As in section 2, a change of variable to get common denominators and integration limits, followed by a series expansion of the numerators, leads to the following expression for the bending angle difference between the two rays:

$$\alpha_1(a) - \alpha(a) = \frac{C}{f_1^2} \frac{d}{da} \int_F N_e ds + \frac{1}{2} \frac{C^2}{f_1^4} a \frac{d^2}{da^2} \int_F N_e^2 ds. \quad (32)$$

A similar expression holds for $\alpha_2(a) - \alpha(a)$, just changing the subscript on the frequency. The residual from the geomagnetic field has been omitted, and the calculations have been carried out to an order concordant with the derivations of the residuals for the phase path correction method in section 2. Applying the bending angle correction (2) gives

$$\alpha_C(a) = \alpha(a) + \frac{1}{2} \frac{C^2}{f_1^2 f_2^2} a \frac{d^2}{da^2} \int_F N_e^2 ds. \quad (33)$$

Obviously, the last term on the right-hand side corresponds to the minor dispersion term. This term is also a generalization of the residual estimated by Vorob'ev and Krasil'nikova [1994] (although its correctness is still dependent on the spherical symmetry assumption). Terms corresponding to the major and the higher-order dispersion residual terms are inherently eliminated using the bending angle correction method (in fact, they are already absent in (32)). Therefore ignoring the minor dispersion term in the improved phase path correction method will give almost the same systematic errors on bending angles as the bending angle correction method does. This was verified by numerical simulations and was found to be true not only in the daytime symmetry case but also in the two asymmetrical cases. This is shown in Figures 10a and 10b, where the bending angle residuals are plotted for the bending angle correction method and the improved phase path correction method (using (26) and (27) with $J = 0$). It is obvious from these plots that the two approaches give very similar results in the range 40–80 km in all cases. Below 40 km, insignificant differences between the two methods occur because of numerical uncertainties and approximations in the calculation of the major term as explained in section 3. Above 80 km, small differences are due to the third- and higher-order dispersion residuals in the phase path correc-

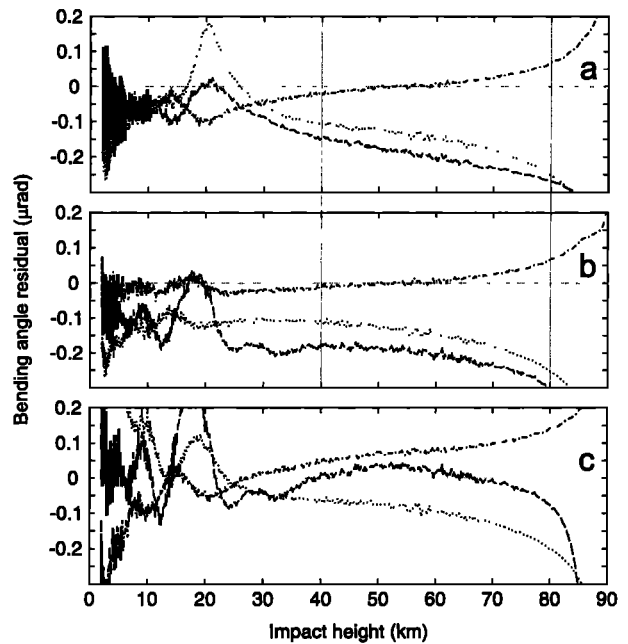


Figure 10. Bending angle residuals versus impact height. (a) Using the bending angle correction method. (b) Using the improved phase path correction method without the minor dispersion term. (c) Using the improved phase path correction method including the minor dispersion term. Dashed curves are for the daytime symmetry case, dash-dotted curves are for day-night asymmetry, and dotted curves are for night-day asymmetry. The thin vertical lines indicate the range 40–80 km where curves in Figures 10a and 10b are very similar.

tion method, which are not present in the bending angle correction method. However, at these altitudes the neutral atmospheric bending angle is so small that the relative error becomes very large beyond 80 km in either method. Figure 10c shows the residuals for the improved phase path correction method when the minor dispersion term is modeled using the model value of the vertical TEC above the occultation site.

These results indicate that the bending angle correction method could be modified to give almost equal results to those of the improved phase path correction method, by estimating the last term in (33). However, the remaining residual, in any case (being on the order of 10^{-7} rad), is so small that noise (being on the order of 10^{-6} – 10^{-5} rad in the GPS/MET experiment) may be the limiting error in practice.

The angular phase correction method 3 may be handled in a similar way, starting from the expressions

$$\int_F N_n ds = 2 \int_{r_F}^{\infty} \frac{N_n \mu_n r dr}{\sqrt{\mu_n^2 r^2 - a^2}}, \quad (34)$$

$$\int_1 N_1 ds = 2 \int_{r_F}^{\infty} \frac{N_1 \mu_1 r dr}{\sqrt{\mu_1^2 r^2 - a_1^2}}. \quad (35)$$

The linear combination (3) is formed at the same altitude h (or perigee radius r_F), and therefore the impact parameters become a little different, resulting in $(a_1 - a)/a \approx -(C/f_1^2)N_e(r_F)$. Almost similar operations to the ones leading to (14) now lead to a similar expression for $\int_1 N_e ds - \int_F N_e ds$, with $\Delta a = -a(C/f_1^2)N_e(r_F)$. A corresponding expression for $\int_1 N_n ds - \int_F N_n ds$ turns out to be totally negligible (basically because comparisons are now made for ray paths having the same perigee altitude). As mentioned in section 1 the angular phase correction is used in connection with the Fresnel diffraction correction [Melbourne *et al.*, 1994; Mortensen and Høeg, 1998], and additional residual terms leak into the correction via the Fresnel deconvolution process and geometrical simplifications. Disregarding such terms and considering only residuals from the geometrical optics approximation, the linear combination of angular phases $\Psi_1(h)$ and $\Psi_2(h)$ then results in

$$\Psi_C(h) = \Psi_F(h) + \frac{C^2(k_1 + k_2)}{f_1^2 f_2^2} \left[a \frac{d}{da} \int_F N_e^2 ds - N_e(h)a \frac{d}{da} \int_F N_e ds \right], \quad (36)$$

$\Psi_F(h) = (k_1 + k_2) \int_F N_n ds$ being the ideal ionosphere-free angular phase and the remaining terms being residual errors. To lowest order, the ionosphere-free bending angle α can be derived from the angular phase by differentiation [Melbourne *et al.*, 1994] (this is a good approximation for a thin atmosphere but is not valid in the lowest part of the Earth's troposphere). Thus comparing the first term in the square brackets in (36) with the last term in (33), we see that the term corresponding to the minor dispersion term has become a factor of 2 larger. Therefore equalizing the bending angle with the derivative of the angular phase (apart from the constant $(k_1 + k_2)$) will introduce a twice as large residual error, compared to the bending angle correction method, due to this term alone (note that $d[a(d/da)]/da \approx a(d^2/da^2)$) because the characteristic scale heights of the atmosphere and ionosphere are orders of magni-

tude smaller than the impact parameter at any altitude). The second term in the square brackets in (36) is due to the small impact parameter separation, and it depends on the S-S TEC gradients as well as the electron density at the perigee altitude. It can probably be ignored in most cases. For the double-Chapman layer model used in the simulations this term was less significant than the third-order dispersion residual ignored in the derivations in section 2.

6. Conclusion

The results presented in this paper do not represent the first attempt in the literature to estimate the dispersion effects on the phase path observables in GPS occultation measurements. However, it has not previously been shown that the first-order dispersion residual from the traditional dual-frequency combination is made up of two parts, a major and a minor, of which the major part is directly related to the vertical gradients of the satellite-to-satellite total electron content (S-S TEC), being available from the occultation measurements themselves. It is also the first time that an estimate of the second-order dispersion effects has been derived and quantified.

The theoretical work here focused on expressing both the L1 and L2 phase paths along the imaginary LF path, using a transformation into the radial domain, assuming spherical symmetry. The results provide a mean to correct the residuals from the traditional dual-frequency combination, thereby obtaining an improved dual-frequency phase path correction method taking into account the dispersion effects.

Within the limits of geometrical optics, this improved phase path correction method is nearly equivalent to the existing, and widely used, bending angle correction method. The main difference between the two methods lies in the minor dispersion term, which contributes to an almost linear phase path trend below the E layer. This trend, if not accounted for, maps into a bending angle residual which is, in fact, equivalent to the main residual obtained with the bending angle correction method.

Unfortunately, estimation of the minor dispersion term needs auxiliary information of, e.g., the vertical TEC above the occultation site. Having that, the minor dispersion term may be evaluated and removed to a large extent. However, this is only true under spherical symmetry conditions. In general, horizontal gradients in the ionosphere limit the correction accuracy, and if the occultation happens to be in a

day-night transition zone, residuals of the order of a few centimeters may still remain using the improved phase path correction method. Such residuals map into bending angle residuals of the order of 10^{-7} rad. Occultations cutting through regions having other sharp horizontal gradients, like the equatorial anomaly or the auroral zone, may give similar residuals.

Sharp vertical gradients at the bottom of the ionosphere attributed to sporadic *E* layers occur frequently and will cause scintillations in the phase and amplitude occultation data [Hajj and Romans, 1998]. Such scintillations, caused by diffraction and multipath propagation and thus violating the geometrical optics assumption, will produce large errors in the ionosphere-corrected phase path. Such effects could be reduced or eliminated using the diffraction correction/back propagation method applied to the ionosphere (M. E. Gorbunov, paper presented at Global Navigation Satellite Systems workshop in Hamburg, 1998), but usually the influence of these disturbances is at such high altitudes that this part of the data may be disregarded for the retrieval of neutral atmospheric parameters.

It should be emphasized that the method of diffraction correction/back propagation [Karayel and Hinson, 1997; Gorbunov and Gurvich, 1998b] cannot be used in conjunction with an ionosphere phase path correction method, since diffraction-corrected bending angles for each of the GPS signals are estimated before ionosphere correction is performed, and then the bending angle correction method is used.

For any ionosphere correction method the mapping of residual bending angle errors into retrieved refractivity, pressure, and temperature highly depends on the technique used for further data processing. Theoretically, the bending angle profile to infinity altitude is required in the integration of bending angles to obtain the refractivity profile. Obviously, this is not available, and in practice, extrapolation and/or use of model bending angles above some altitude is necessary in order to get accurate and useful refractivities at the highest altitudes.

Measurement noise is yet another obstacle in practice, and indeed a serious one. Some kind of filtering of the phase-time series is needed before further processing into Doppler shifts and bending angles. Still, bending angles will inevitably have random fluctuations which blow up the relative error at some high altitude and above. This calls for statistical optimization techniques where a model bending angle profile gradually replaces the data profile above

some specified altitude depending on the noise level [Sokolovskiy and Hunt, 1996]. For the data available from the GPS/MET experiment an altitude of 40–50 km seems to be appropriate [Hocke, 1997; Gorbunov and Gurvich, 1998a]; an altitude much lower than the altitude where ionosphere residuals become significant. However, future instrumentation will probably supply data of a better quality and reduced noise level [Kursinski et al., 1997], allowing the statistical optimization level to be raised into the height region where ionosphere residuals may become the limiting error source.

It is hoped that this paper may contribute to a better understanding of the systematic errors in GPS radio occultation measurements arising as a consequence of insufficient calibration of ionospheric effects and that this understanding may foster new ideas for the development of efficient ionosphere correction methods.

Appendix A: Series Expansion of the Numerators in (10) and (12)

Generally, expressions for $r(x)$ are needed, which may be obtained by rearranging the equation defining the change of variable leading to (10) and (12):

$$r(x) = \left(1 + \frac{\Delta a}{a}\right) \frac{x}{\mu[r(x)]}, \quad (\text{A1})$$

with $\Delta a = a_1 - a$ and $\mu = \mu_1$ in the case of γ_1 and $\Delta a = 0$ and $\mu = \mu_n$ in the case of γ_F . Equation (A1) is an implicit equation, with the sensitivity of the right-hand side to $r(x)$ through μ being rather weak. Using an iterative process, it is possible to calculate a series expansion of r as a function of x . To first order in N and $\Delta a/a$, we get

$$r(x) = x - x \left(N - \frac{\Delta a}{a} \right), \quad (\text{A2})$$

where $N = N(x)$ is to be thought of as the refractivity functional dependence of the radial distance, but with the argument r replaced by x . Using (A2), a Taylor series expansion to the second order now gives an expression for the electron density,

$$N_e[r(x)] = N_e - x \left(N - \frac{\Delta a}{a} \right) \frac{dN_e}{dx} + \frac{1}{2} \left(\frac{\Delta a}{a} \right)^2 x^2 \frac{d^2 N_e}{dx^2}, \quad (\text{A3})$$

omitting terms of the second order containing N . Here, like before, $N_e = N_e(x)$ on the right-hand side denotes the electron density functional dependence of the radial distance with the argument r replaced by x .

The order to which the expansions in (A2) and (A3) are carried out is not obvious and will be given some attention here. First of all, it must be based on the contribution of each term to the resulting order of magnitude when terms are combined and evaluated through integrals like (10) and (12). Second, the relative order of magnitude of terms depending on the refractivity N is totally different from the relative order of magnitude of terms depending on the electron density N_e . For example, in the lower troposphere, terms like $x(d^2N/dx^2)$, $x^2(d^3N^2/dx^3)$, $x^3(d^4N^3/dx^4)$, ..., contribute with almost equal magnitude in an integral evaluation weighted with the kernel $(x^2 - a^2)^{-1/2}$, whereas in the ionosphere the contributions from the same sequence, with N_e replacing N , fall off quite rapidly. Also, each term varies by several orders of magnitude at different heights (for different values of the impact parameter) in the neutral atmosphere and ionosphere. All this makes the order of magnitude evaluation difficult to handle in a general sense. The evaluation of which terms to include and which to ignore was therefore based on careful estimation of the final contribution from each term at all heights (surface to ~ 110 km) through numerical computations. In the computations, representative models of the neutral atmosphere and of the ionosphere at daytime, solar maximum, were used. Terms neglected in (A2) and (A3), and in the following derivations, turned out to contribute at least 1 order of magnitude less than terms included. To avoid too many unnecessary terms, which will rapidly expand the derivations, only terms needed to finally achieve the goal of estimating $\gamma_1 - \gamma_F$ to second order will be written.

To the required order, it now follows that

$$\begin{aligned} x \frac{dr}{dx} N_e[r(x)] &= xN_e - x^2 \frac{d}{dx} (N_e N) + \frac{\Delta a}{a} x \frac{d}{dx} (xN_e) \\ &+ \frac{1}{2} \left(\frac{\Delta a}{a} \right)^2 x^3 \frac{d^2 N_e}{dx^2}. \end{aligned} \quad (\text{A4})$$

In the case of γ_F ($N = N_n$, $\Delta a = 0$), all terms but the first on the right-hand side of (A4) vanish. The second term vanishes because $N_e N_n$ is practically negligible anywhere in the atmosphere. In the case of

γ_1 all terms are needed, and in the second term, $N_e N = N_e [N_n + N_i(f_1)] \approx -(C/f_1^2) N_e^2$. Inserting these results into (10) and (12) and doing the subtraction, we obtain (13).

Appendix B: Some Basic Relations

In this appendix the following three approximate relations are derived:

$$a \frac{d}{da} \int_F N_e ds = 2 \int_a^\infty \frac{x \frac{d(xN_e)}{dx} dx}{\sqrt{x^2 - a^2}}, \quad (\text{B1})$$

$$a \frac{d}{da} \int_F N_e^2 ds = 2 \int_a^\infty \frac{x^2 \frac{dN_e^2}{dx} dx}{\sqrt{x^2 - a^2}}, \quad (\text{B2})$$

$$a^2 \frac{d^2}{da^2} \int_F N_e ds = 2 \int_a^\infty \frac{x^3 \frac{d^2 N_e}{dx^2} dx}{\sqrt{x^2 - a^2}}. \quad (\text{B3})$$

First, we note that to sufficient order,

$$\int_F N_e ds = 2 \int_a^\infty \frac{xN_e dx}{\sqrt{x^2 - a^2}}, \quad (\text{B4})$$

which follows from (10), (A4), and the definition of γ_F . Likewise, we may write

$$\int_F N_e^2 ds = 2 \int_a^\infty \frac{xN_e^2 dx}{\sqrt{x^2 - a^2}}. \quad (\text{B5})$$

Using Leibnitz's rule for differentiation of integrals [e.g., *Spiegel*, 1968], we have

$$\begin{aligned} \frac{d}{da} \int_b^\infty \frac{xN_e dx}{\sqrt{x^2 - a^2}} &= \int_b^\infty \frac{d}{da} \left(\frac{xN_e}{\sqrt{x^2 - a^2}} \right) dx - \frac{bN_e(b)}{\sqrt{b^2 - a^2}} \frac{db}{da} \\ &= \int_b^\infty \frac{axN_e dx}{(x^2 - a^2)^{3/2}} - \frac{bN_e(b)}{\sqrt{b^2 - a^2}}, \end{aligned}$$

where we have used $(xN_e/\sqrt{x^2 - a^2}) \rightarrow 0$ at the upper limit. The parameter $b = a + \varepsilon$ (ε is small and positive) is introduced to avoid singularity problems at the lower limit. Integrating the first term on the right-hand side by parts, we further get

$$\frac{d}{da} \int_b^\infty \frac{x N_e dx}{\sqrt{x^2 - a^2}} = \int_b^\infty \frac{a \frac{dN_e}{dx} dx}{\sqrt{x^2 - a^2}} - \frac{(b-a) N_e(b)}{\sqrt{b^2 - a^2}},$$

which in the limit $\varepsilon \rightarrow 0$ gives

$$\frac{d}{da} \int_a^\infty \frac{x N_e dx}{\sqrt{x^2 - a^2}} = \int_a^\infty \frac{a \frac{dN_e}{dx} dx}{\sqrt{x^2 - a^2}}. \quad (\text{B6})$$

If we make the plausible assumption that $N_e(x)$, for large values of x , falls off more rapidly than linear, it is easily shown, again using integration by parts, that

$$\int_a^\infty \frac{x \frac{d(xN_e)}{dx} dx}{\sqrt{x^2 - a^2}} - \int_a^\infty \frac{a^2 \frac{dN_e}{dx} dx}{\sqrt{x^2 - a^2}} = 0. \quad (\text{B7})$$

Combining (B4), (B6), and (B7) then gives (B1).

Using (B5), (B6), and (B7) and replacing N_e by N_e^2 in the two latter equations, we see, after a few manipulations, that

$$2 \int_a^\infty \frac{x^2 \frac{dN_e^2}{dx} dx}{\sqrt{x^2 - a^2}} = a \frac{d}{da} \int_F N_e^2 ds - \int_F N_e^2 ds. \quad (\text{B8})$$

For the ionosphere conditions that were used to estimate the order of magnitude the second term on the right-hand side of (B8) came out more than 1 order of magnitude less than the first term (basically because the characteristic scale heights of the ionospheric electron density are much smaller than the impact parameter), and when substituted into (13), it should be neglected when compared to terms already ignored in the series expansions leading to (13). Neglecting this term, (B8) reduces to (B2).

From the result (B6), using Leibnitz's rule and integration by parts, we find

$$\frac{d^2}{da^2} \int_a^\infty \frac{x N_e dx}{\sqrt{x^2 - a^2}} = \int_a^\infty \frac{d}{dx} \left(x \frac{dN_e}{dx} \right) \frac{dx}{\sqrt{x^2 - a^2}}. \quad (\text{B9})$$

Again, with some manipulations using integration by parts, it is possible to show that

$$\begin{aligned} \int_a^\infty \frac{x^3 \frac{d^2 N_e}{dx^2} dx}{\sqrt{x^2 - a^2}} - \int_a^\infty \frac{a^2 \frac{d}{dx} \left(x \frac{dN_e}{dx} \right) dx}{\sqrt{x^2 - a^2}} \\ = -2 \int_a^\infty \frac{x^2 \frac{dN_e}{dx} dx}{\sqrt{x^2 - a^2}}. \end{aligned} \quad (\text{B10})$$

Combining (B4), (B9), (B10), and (B8) and using N_e instead of N_e^2 in the latter, we arrive at

$$\begin{aligned} \int_a^\infty \frac{x^3 \frac{d^2 N_e}{dx^2} dx}{\sqrt{x^2 - a^2}} = \frac{1}{2} a^2 \frac{d^2}{da^2} \int_F N_e ds - a \frac{d}{da} \int_F N_e ds \\ + \int_F N_e ds. \end{aligned} \quad (\text{B11})$$

Again, numerical computations showed that the first term on the right-hand side of (B11) is far the largest and is the only term that should be taken into account when compared to other terms in (13). Neglecting the last two terms in (B11), the equation reduces to (B3).

Appendix C: Derivation of $\Delta\alpha$

From the occultation geometry (see Figure 2) we may get a relation between the bending angle α_1 and the impact parameter a_1 connected to the L1 path:

$$\alpha_1 = \Theta - \arccos\left(\frac{a_1}{r_L}\right) - \arccos\left(\frac{a_1}{r_G}\right), \quad (\text{C1})$$

where Θ is the angle between the satellite radial vectors. The same relation holds for α and a connected to the LF path in Figure 2. Doing a Taylor expansion around the impact parameter, we can then calculate a relation between $\Delta\alpha = \alpha_1 - \alpha$ and Δa , which to first order becomes

$$\Delta\alpha = \Delta a \left(\frac{1}{D_L} + \frac{1}{D_G} \right), \quad (\text{C2})$$

where $D_L = \sqrt{r_L^2 - a^2}$ and $D_G = \sqrt{r_G^2 - a^2}$. The second-order term proportional to $(\Delta a)^2$ (not written in (C2)) turns out to be more than 4 orders of magnitude less than the first-order term and is not needed here. The bending angle difference can be obtained in yet another way using the integral form for the bending angles [Fjeldbo et al., 1971]:

$$\alpha = -2a \int_{r_f}^{\infty} \frac{(d \ln \mu_n/dr) dr}{\sqrt{\mu_n^2 r^2 - a^2}}, \quad (C3)$$

$$\alpha_1 = -2a_1 \int_{r_1}^{\infty} \frac{(d \ln \mu_1/dr) dr}{\sqrt{\mu_1^2 r^2 - a_1^2}}. \quad (C4)$$

Doing similar operations on (C3) and (C4) to those done in going from (9) and (11) to (14), it is possible to find

$$\Delta\alpha = \frac{C}{f_1^2} \frac{d}{da} \int_F N_e ds + \Delta\alpha \left(\frac{C}{f_1^2} \frac{d^2}{da^2} \int_F N_e ds + \frac{d\alpha}{da} \right). \quad (C5)$$

Combining (C2) and (C5), we obtain

$$\begin{aligned} \Delta a \left[\frac{1}{D} \left(1 - D \frac{d\alpha}{da} \right) - \frac{C}{f_1^2} \frac{d^2}{da^2} \int_F N_e ds \right] \\ = \frac{C}{f_1^2} \frac{d}{da} \int_F N_e ds, \end{aligned} \quad (C6)$$

where the reduced distance D has been defined as

$$\frac{1}{D} = \frac{1}{D_L} + \frac{1}{D_G}. \quad (C7)$$

The first term in the square brackets on the left-hand side of (C6) was estimated to be at least 1 order of magnitude larger than the second term. Introducing the defocusing factor,

$$\zeta = \left(1 - D \frac{d\alpha}{da} \right)^{-1}, \quad (C8)$$

and rearranging to the order needed for insertion in (14), we then get

$$\Delta a = D\zeta \frac{C}{f_1^2} \frac{d}{da} \int_F N_e ds \left[1 + D\zeta \frac{C}{f_1^2} \frac{d^2}{da^2} \int_F N_e ds \right]. \quad (C9)$$

It is interesting to notice here that the product $D\zeta$ is practically proportional to the square of the radius of the first Fresnel zone of the signal beam in the vertical direction [e.g., *Melbourne et al.*, 1994].

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Erratum

In the originally published version of this article, Equation 33 contained a typographical error. The error has since been corrected and this version may be considered the authoritative version of record.

In Equation 33, the plus "+" sign was changed to minus "-".

$$\alpha_c(a) = \alpha(a) - \frac{1}{2} \frac{C^2}{f_1^2 f_2^2} a \frac{d^2}{da^2} \int_F N_\epsilon^2 ds. \quad (33)$$